We can also see that all reversible machines must lift to *exactly the same height.* Suppose that B was really reversible also. The argument that *Y* is not higher than *X* is, of course, just as good as it was before, but we can also make our argument the other way around, using the machines in the opposite order, and prove that *X is not higher than Y.* This, then, is a very remarkable observation because it permits us to analyze the height to which different machines are going to lift something *without looking at the interior mechanism.* We know at once that if somebody makes an enormously elaborate series of levers that lift three units a certain distance by lowering one unit by one unit distance, and we compare it with a simple lever which does the same thing and is fundamentally reversible, his machine will lift it no higher, but perhaps less high. If his machine is reversible, we also know exactly *how* high it will lift. To summarize: every reversible machine, no matter how it operates, which drops one pound one foot and lifts a three-pound weight always lifts it the same distance, *X*. This is clearly a universal law of great utility. The next question is, of course, what is *X?*

Suppose we have a reversible machine which is going to lift this distance *X,* three for one. We set up three balls in a rack which does not move, as shown in Fig. 4-2. One ball is held on a stage at a distance one foot above the ground. The machine can lift three balls, lowering one by a distance 1. Now, we have arranged that the platform which holds three balls has a floor and two shelves, exactly spaced at distance *X,* and further, that the rack which holds the balls is spaced at distance *X,* (a). First we roll the balls horizontally from the rack to the shelves, (b), and we suppose that this takes no energy because we do not change the height. The reversible machine then operates: it lowers the single ball to the floor, and it lifts the rack a distance *X,* (c). Now we have ingeniously arranged the rack so that these balls are again even with the platforms. Thus we unload the balls onto the rack, (d); having unloaded the balls, we can restore the machine to its original condition. Now we have three balls on the upper three shelves and one at the bottom. But the strange thing is that, in a certain way of speaking, we have not lifted *two* of them at all because, after all, there were balls on shelves 2 and 3 before. The resulting effect has been to lift *one ball* a distance 3*X*. Now, if 3*X* exceeds one foot, then we can *lower* the ball to return the machine to the initial condition, (f), and we can run the apparatus again. Therefore 3*X* cannot exceed one foot, for if 3*X* exceeds one foot we can make perpetual motion. Likewise, we can prove that *one foot cannot exceed* 3*X*, by making the whole machine run the opposite way, since it is a reversible machine. Therefore 3*X* is neither *greater nor less than a foot,* and we discover then, by argument alone, the law that *X* = ⅓ foot. The generalization is clear: one pound falls a certain distance in